



Filter Wizard

2134

Created on 02/23/2026



Filter Wizard Design Report

Filter Requirements for Band-Pass, 4th order Chebyshev

Specifications: Optimize: Specific Parts; +Vs: 3.3; -Vs: 0

Gain: 40 dB

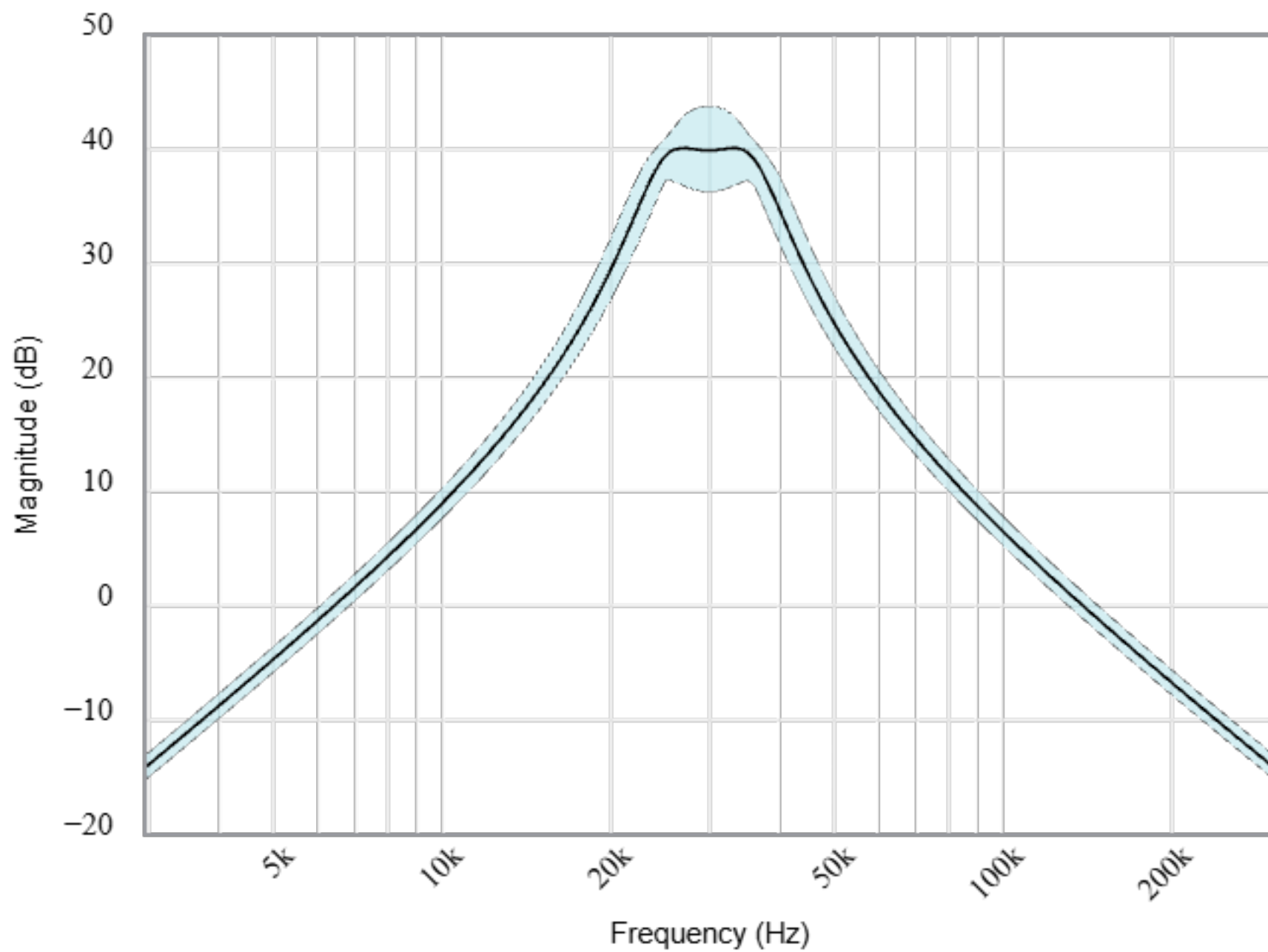
Passband: -3dB at 15kHz

Stopband: -40dB at 150kHz

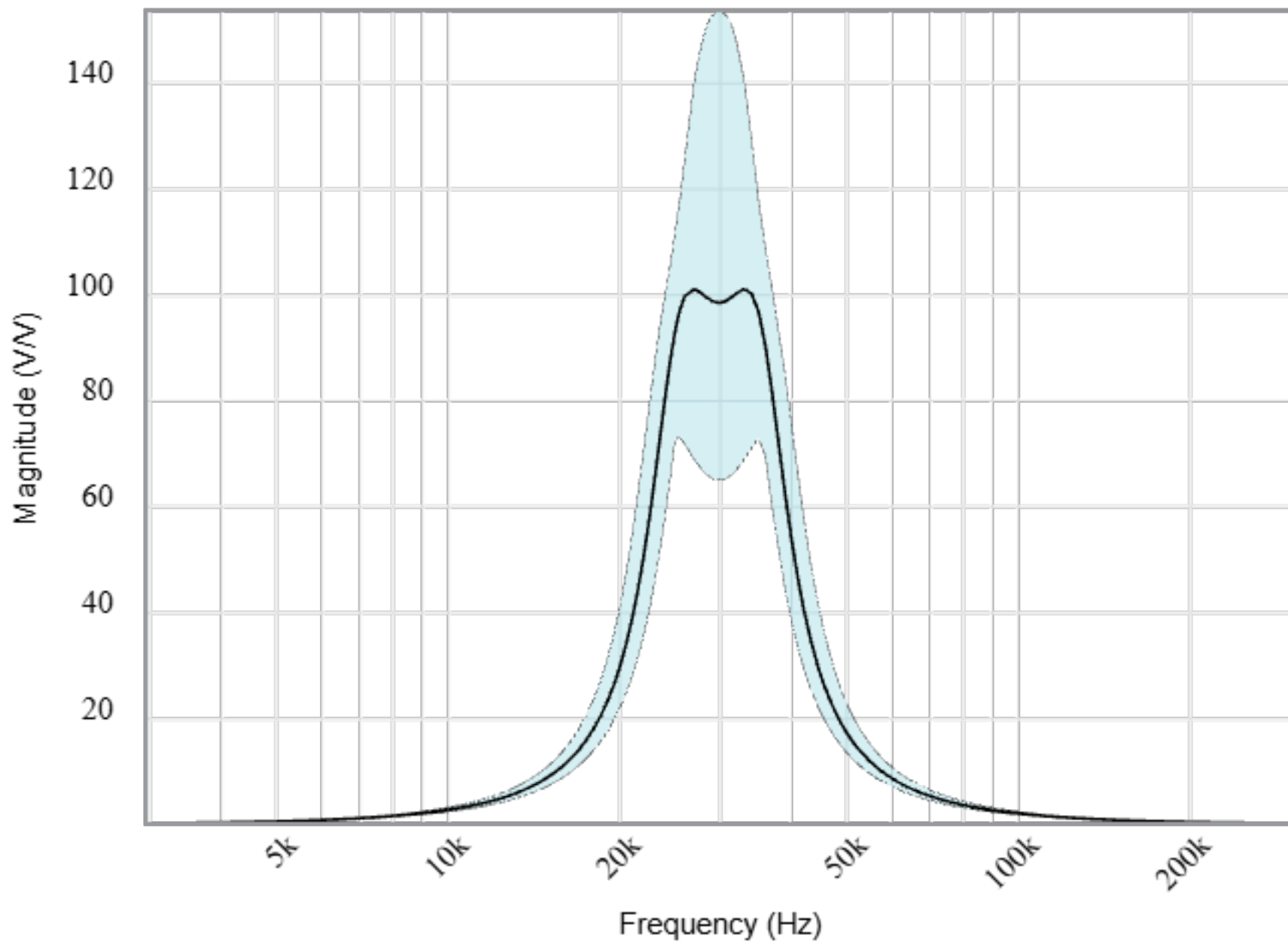
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

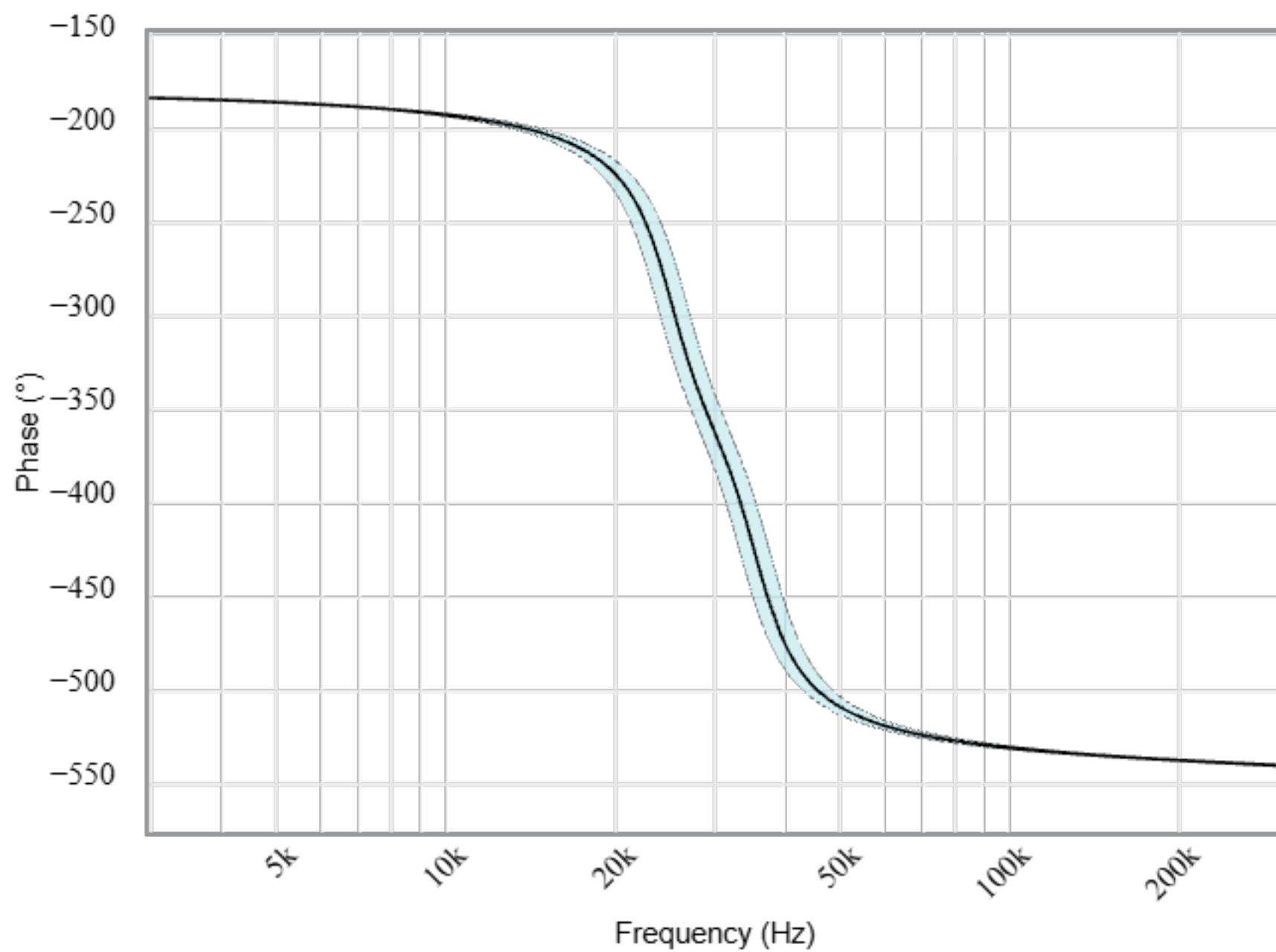
Magnitude(dB)



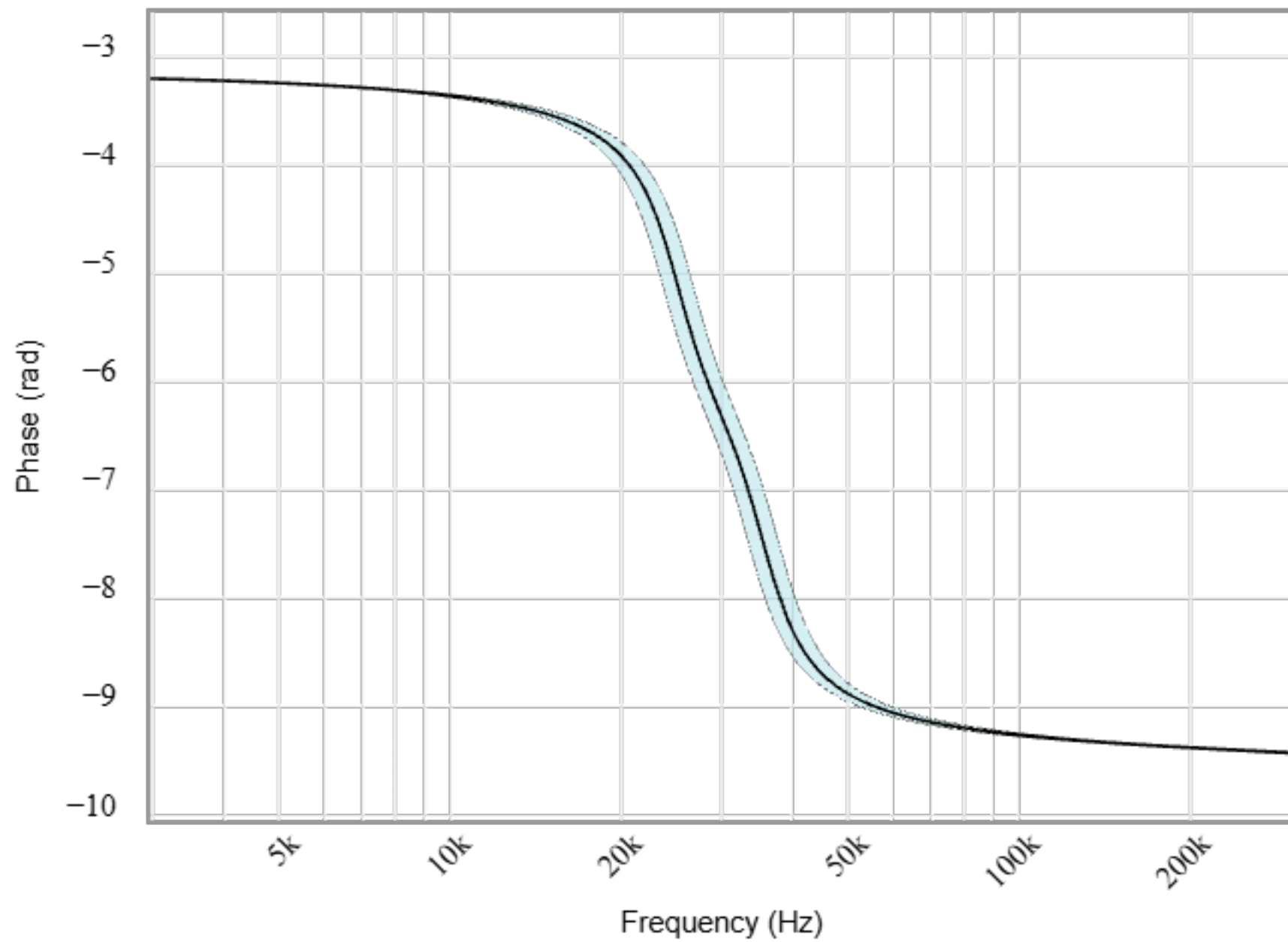
Magnitude(Volts per Volt)



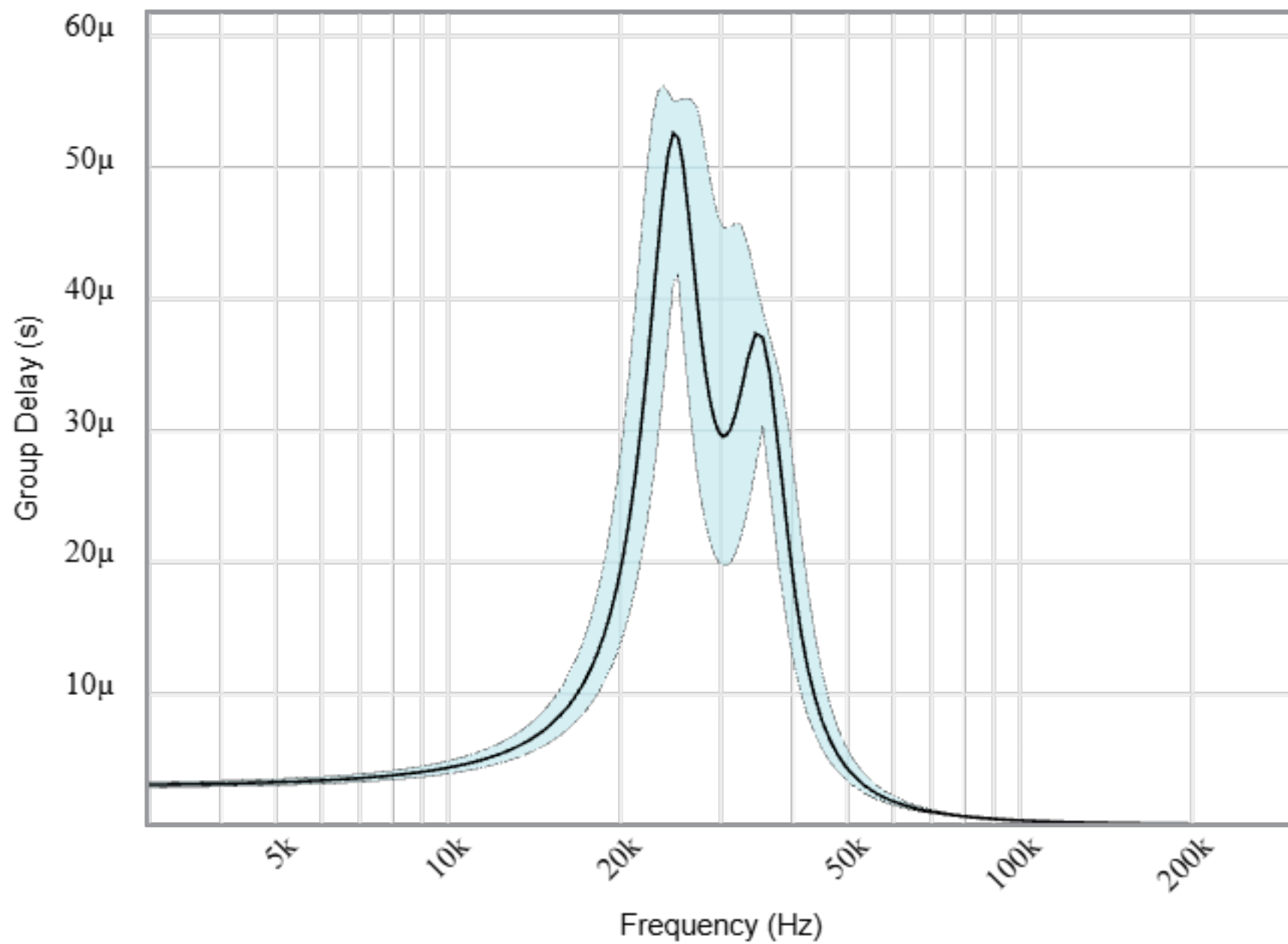
Phase(degrees)



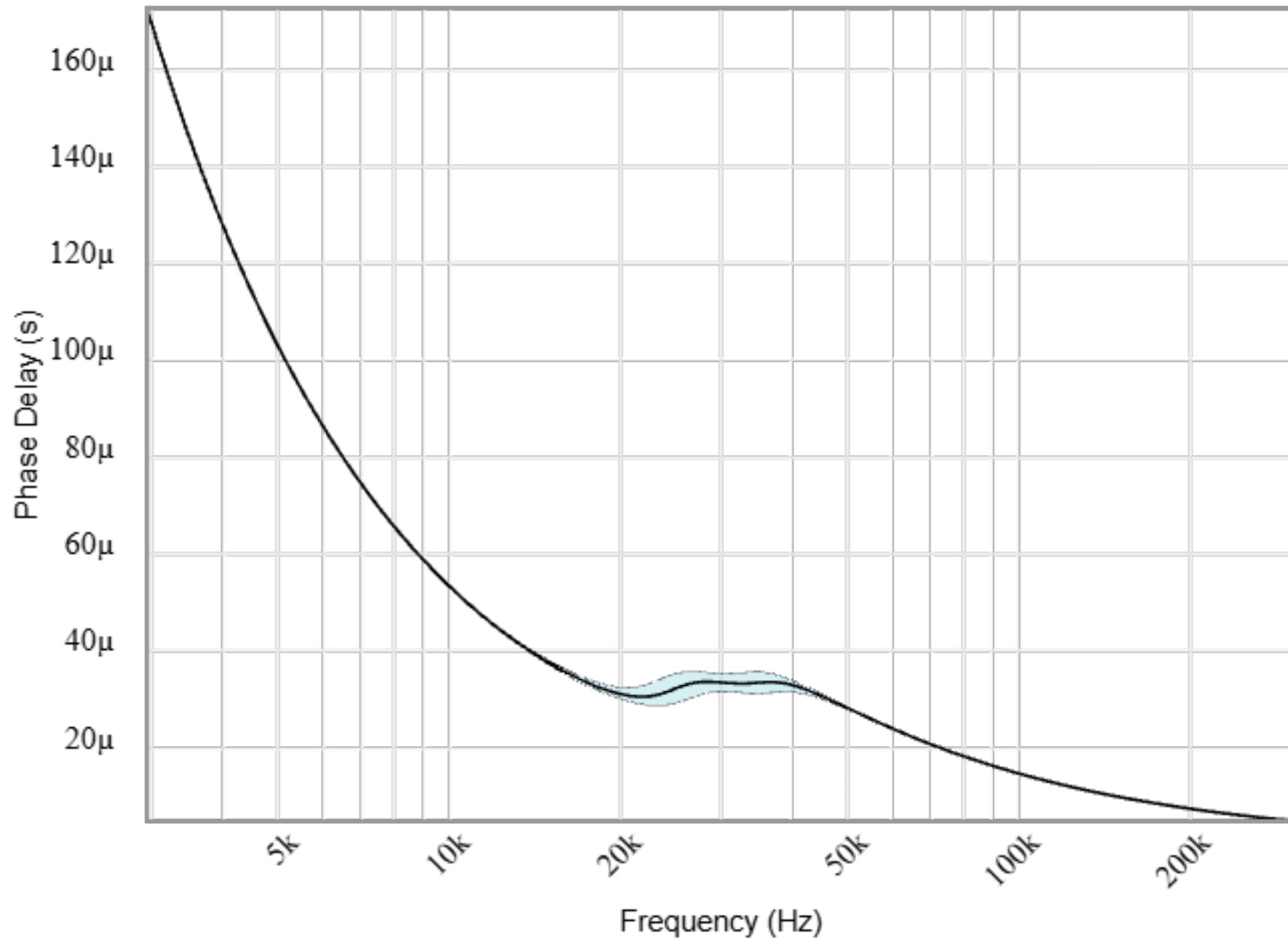
Phase(radians)



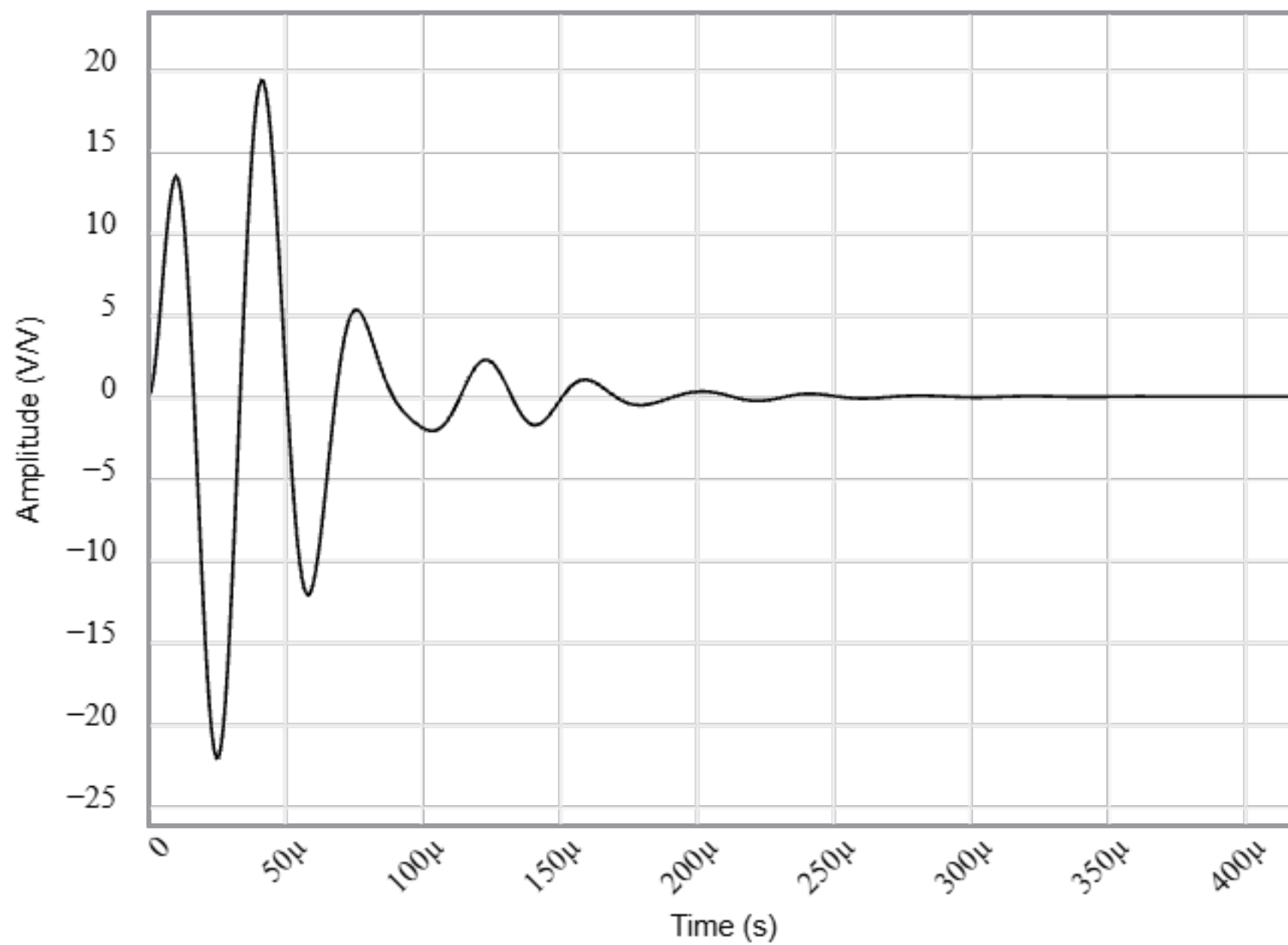
Group Delay



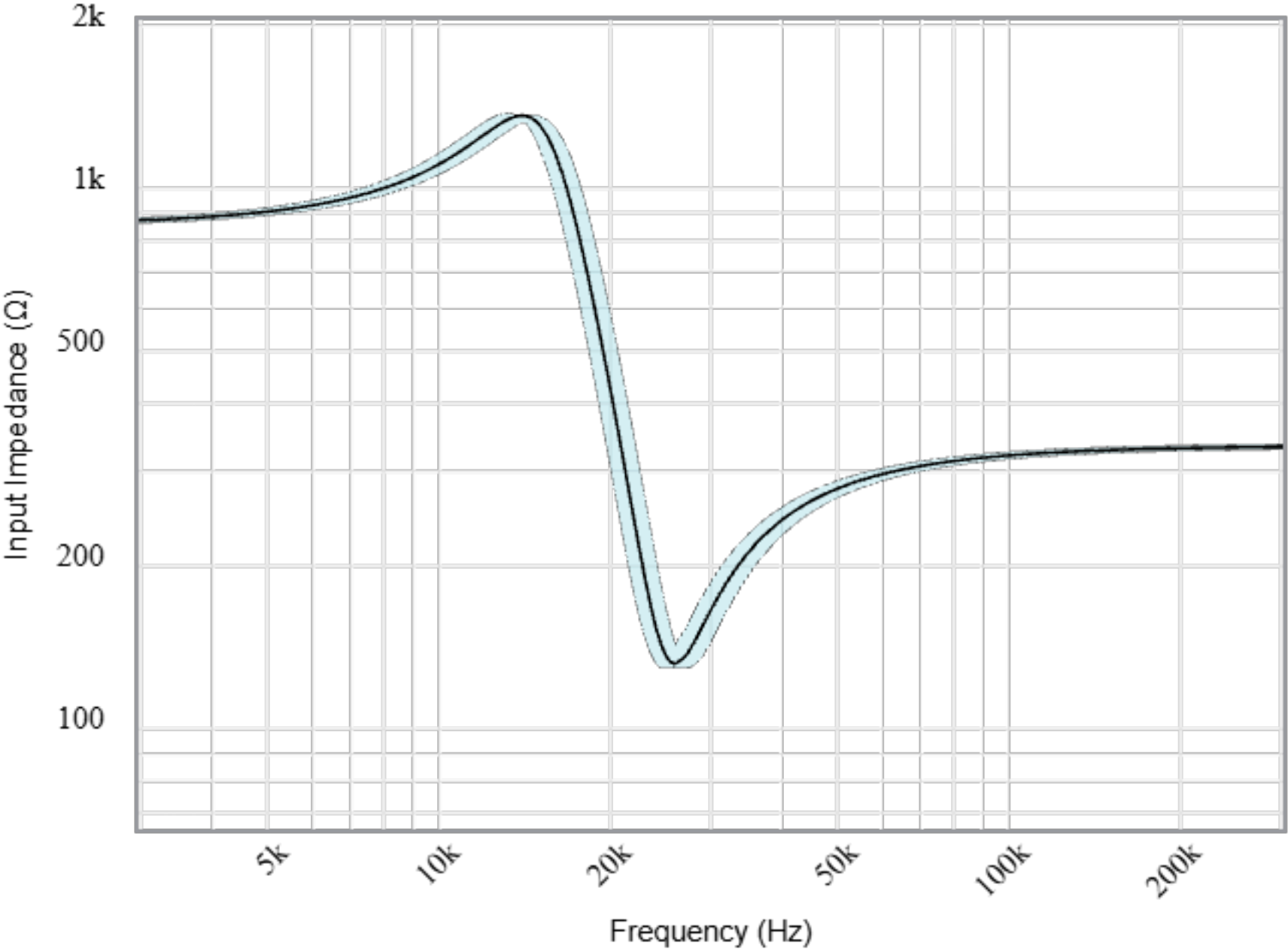
Phase Delay



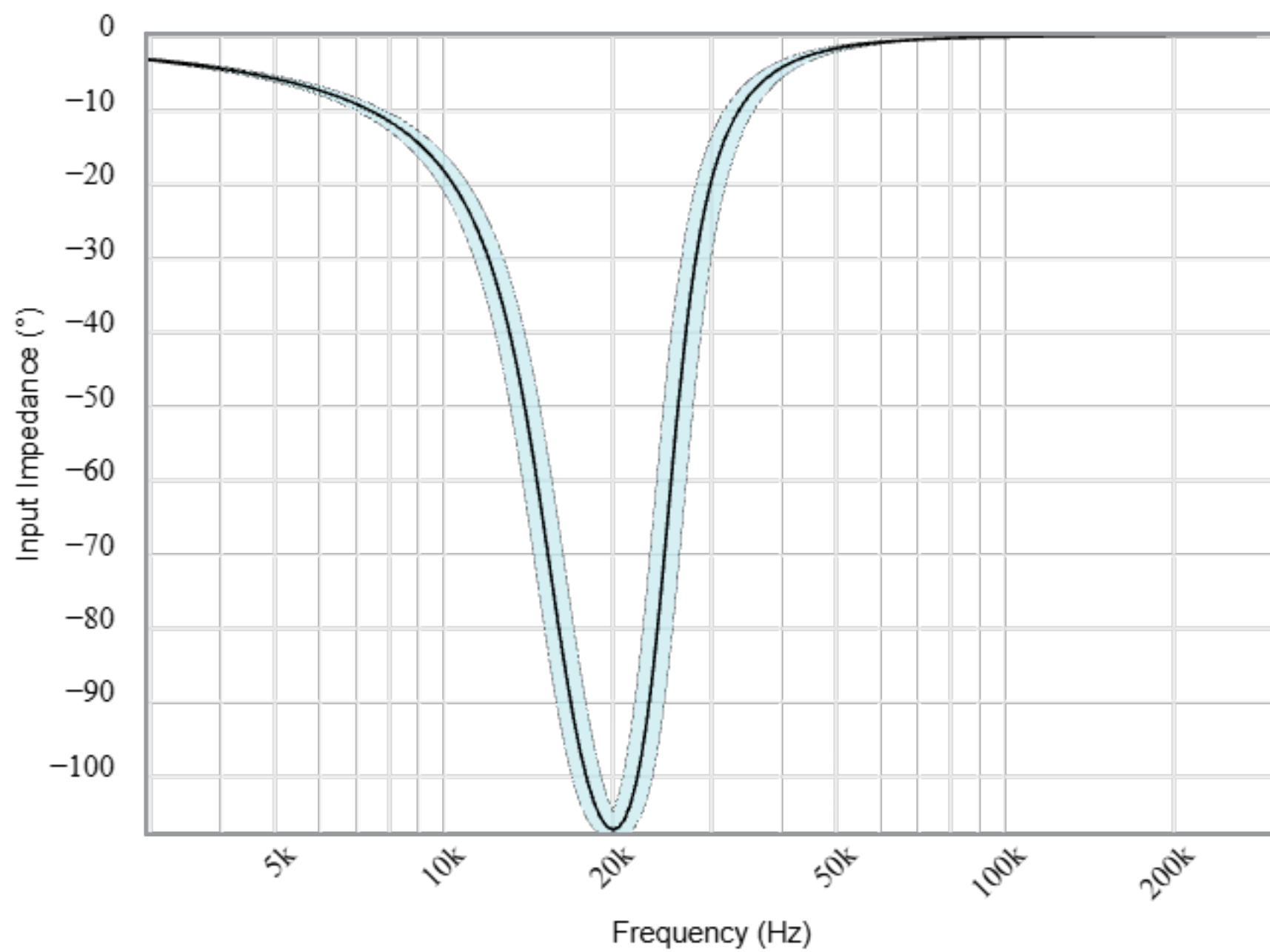
Step Response



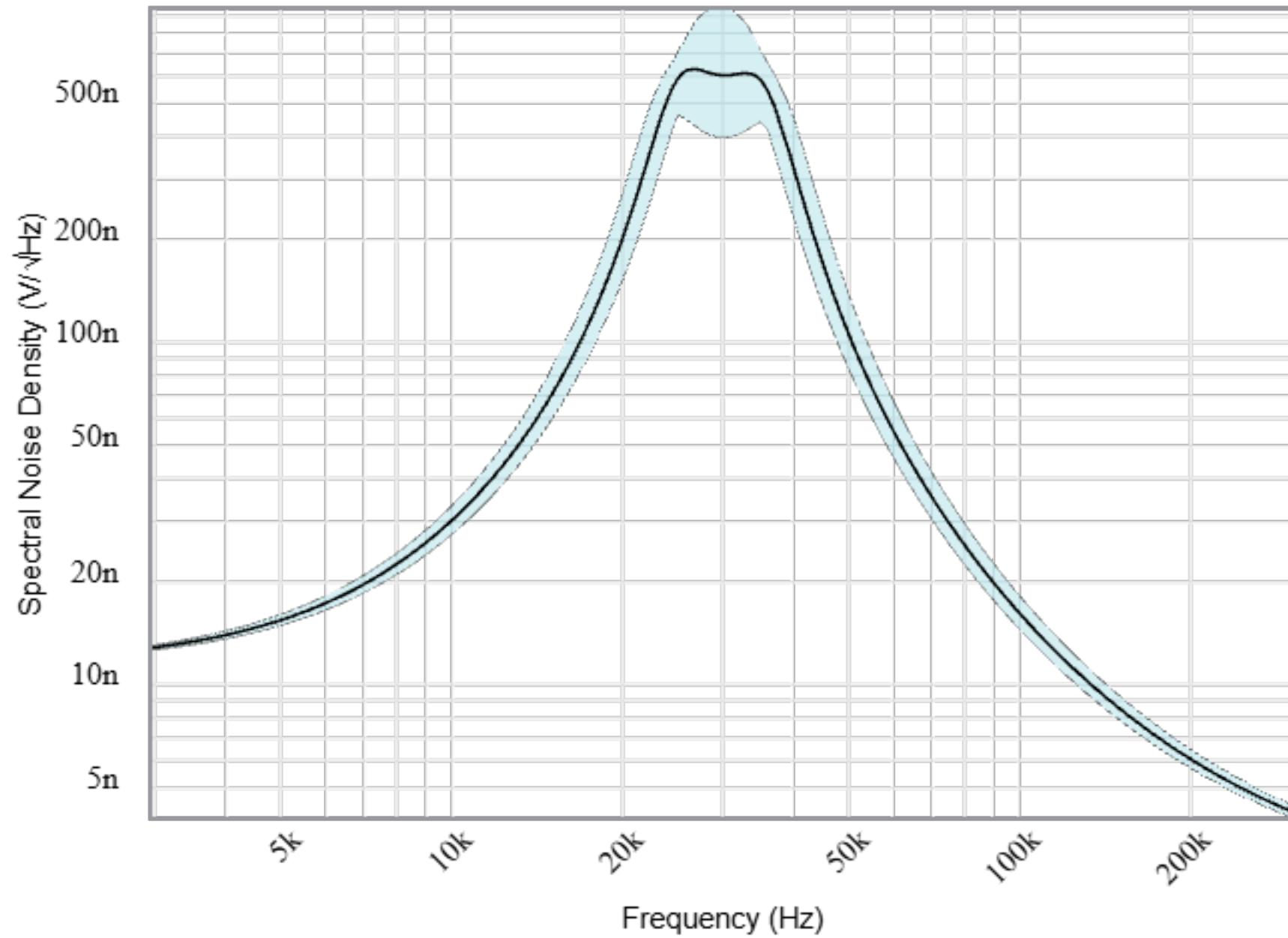
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 2 op amp stage(s) with the following characteristics



2nd order
Band-Pass
Multiple
Feedback

Target

Simulated

Gain (V/V):

16.3

15.8 to 16.5

f_p (Hz):

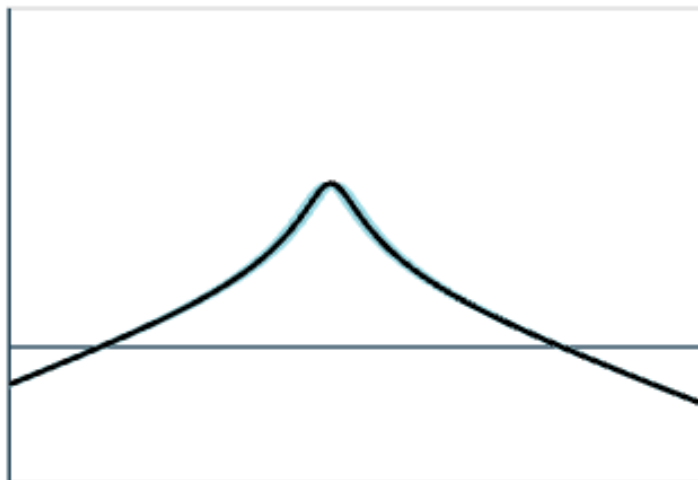
25.2k

23.6k to 26.6k

Q:

3.63

3.61 to 3.69



2nd order
Band-Pass
Multiple
Feedback

Target

Simulated

16.3

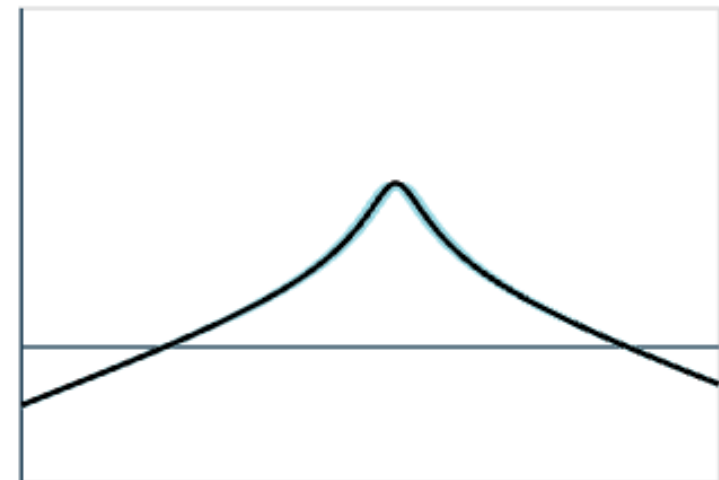
15.9 to 16.6

35.8k

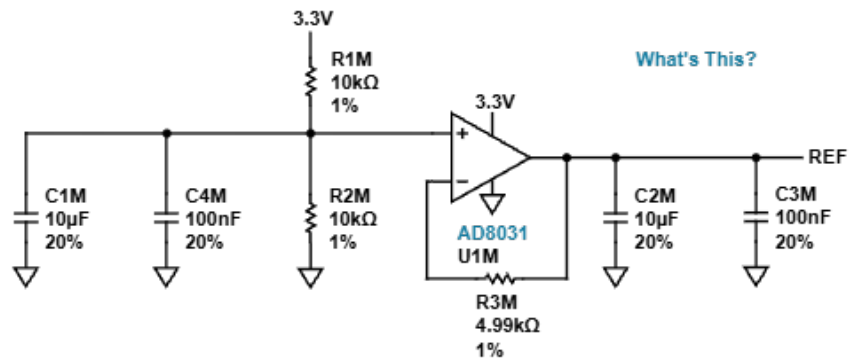
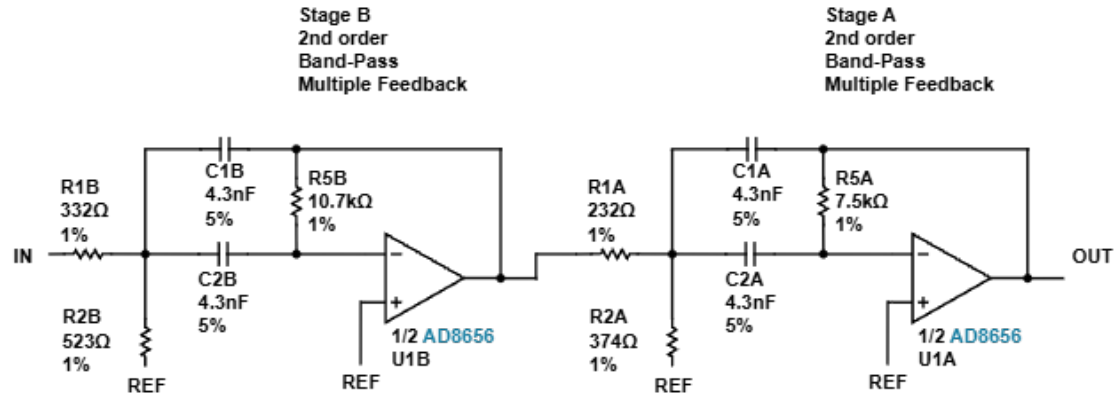
33.5k to 37.8k

3.63

3.61 to 3.69



Circuit



BYPASS CAPACITORS

